Prachee Gupta

pg2871@nyu.edu | LinkedIn | GitHub

Experience

SDE Intern Jun 2026 – Aug 2026
Amazon Web Services (AWS) *USA*

- Designed and implemented a two-phase video color-space derivation engine for AWS Elemental MediaConvert, supporting major customers including Amazon Prime Video, addressing 216M+ warnings across 11K+ unique accounts over 90 days and preventing incorrect or oversaturated colors in delivered video output.
- Built a standards-based deterministic pipeline (Phase 1) using ITU/SMPTE lookup tables and an aggressive heuristic derivation pipeline (Phase 2) requiring customer opt-in, leveraging resolution, bit depth, and frame properties; added a new console UI option for Phase 2 opt-in and deployed the deterministic pipeline to production.
- Added Hindi language burn-in caption support and resolved localization edge cases involving font layout and character shaping, improving accuracy of Hindi text rendering in localized video output.
- Owned end-to-end development from architecture and design through implementation, testing, reviews, and cross-team integration; authored 2 design docs and 5 PRs, contributing 2,800+ LOC and 122+ tests (86 unit, 36 integration) and fixing 4 bugs.

Software Engineer, Supply Chain Jun 2023 – Aug 2025
Lowe's *Bengaluru, India*

- Designed and implemented role-based access control using Redis with PostgreSQL fallback, restricting administrative operations by role, reducing database calls, and enabling permission updates within seconds for 500+ concurrent users.
- Modernized legacy C-based yard operations by decoupling cross-service workflows and redesigning them in Java/Spring Boot with Kafka, reducing processing time from 60+ seconds to under 30 ms while reducing API timeout errors.
- Led the transition of Bulk Distribution Center (BDC) operations from Blue Yonder to the yard management platform; authored design docs, evolved APIs, and coordinated integration across multiple teams while preserving a node-agnostic architecture.
- Diagnosed and resolved critical production issues during peak sales cycles, ensuring zero operational disruption during high-volume periods.
- Recognized with three Spot Awards and an Individual Excellence Award for innovation, leadership, and reliability under pressure.

Associate Software Engineer, Supply Chain Aug 2021 – Jun 2023
Lowe's *Bengaluru, India*

- Built full-stack features end-to-end, including API handling (Kotlin, Micronaut), UI development (React.js), and DevOps tooling (Docker, Kubernetes, Jenkins, GitLab CI/CD, Prometheus, Grafana, Kibana).
- Engineered a first-of-its-kind yard visualization feature that automated Excel-to-UI layout synchronization, eliminating manual reconciliation and reducing turnaround from days to minutes.
- Managed monitoring platforms including Grafana, Kibana, and Argo CD; worked with SQL and NoSQL databases and implemented Pub/Sub and Coroutines to improve user experience and response times.

Teaching Assistant, Principles of Database Systems Jan 2026 – Dec 2026
NYU *USA*

- Mentored 60+ graduate students on SQL, relational schema design, indexing, query optimization, and transaction management; supported labs and evaluated projects and assignments.

Projects

AI Approach for Autonomous Vehicles to Defend from Adversarial Attacks Jan 2021 – Jan 2022

- Developed an adversarial-defense framework using computer vision and machine learning, achieving 93.56% accuracy; published and presented at a Springer international conference, featured in Algorithms for Intelligent Systems.

Skills

Backend & Distributed Systems: Java, Kotlin, Spring Boot, Micronaut, REST APIs, WebSockets, Apache Kafka

Cloud & Infrastructure: AWS, Docker, Kubernetes, Jenkins, ArgoCD, Grafana, Kibana

Databases & Search: PostgreSQL, Redis, Elasticsearch

Languages: Python, JavaScript, SQL, C++

Education

New York University USA
Master of Science in Computer Engineering. *May 2027*

Indian Institute of Information Technology Nagpur (IIITN) India
Bachelor of Technology in Electronics and Communications. Graduated with Distinction. GPA: 8.64/10. *May 2021*